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Wigner distribution moments associated with the linear canonical transform

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The moments associated with the linear canonical transform is investigated. First, the relationship between the Wigner distribution and the linear canonical transform is reviewed, then the moments associated with the linear canonical transform is investigated in detail, and a similar relation is obtained for the two-dimensional cases. The simulation experiments are also proposed to verify the derived results.

Keywords: linear canonical transform series; Wigner distribution; moments

1. Introduction

Moments and functions of moments have been applied in pattern recognition (Hu 1962; Belkasim, Shridhar, and Ahmadi 1991; Flusser and Suk 1993), edge detection (Liu and Tsai 1990; Luo, Hamitouche, Dillenseger, and Coatrieux 1993), image segmentation (Yokoya and Levine 1989), texture analysis Tuceryan (1994) and other domains of image analysis (Teh and Chin 1988; Teague 1980) and computer vision (Abu-Mostafa and Psaltis 1984; Lo and Don 1989). Often the application of the different mixed time-frequency distributions depend on how informative their moments are and how easily they can be measured or calculated. Moments in the fractional Fourier transform (FRFT) domain have been studied in Bastiaans and Alieva (2002) and Alieva and Bastiaans (2000). The LCT is a linear integral transformation of a four-parameter function and it can be considered as the generalisation of the FRFT and the Fourier transform (FT). Comparing to the FRFT and the FT, the LCT is shown to be more flexible for signal processing. With the research of the LCT, many concepts have been developed in the LCT domain, such as the relationship of the Wigner distribution (the ambiguity function (AF)) in the time-frequency representations has been investigated in Che, Li, and Xu (2012) and Bai, Li, and Cheng (2012), the Properties in Moshinsky and Quesne (1971) and the sampling theorem in Li, Tao, and Wang (2007), Tao, Li, and Wang (2008), Wei, Ran, and Li (2011) and Li and Xu (2012). However, there are no papers published in the moments associated with the LCT. Therefore, it is worthwhile and interesting to investigate it.

Focussing on this problem, this article investigates the Wigner moments associated with the LCT. This article is organised as follows: Section 2 is the preliminary; Section 3 gives the Wigner moments associated with the LCT and Section 4 investigates the similar

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results in the two-dimensional LCT domain. Section 5 discussed the simulation results and the conclusions are given in Section 6.

2. Preliminary

2.1. The linear canonical transform (LCT)

The LCT with parameter $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of a signal $f(t)$ is defined as (Moshinsky and Quesne 1971; Bai et al. 2012)

$$F_A(u) = L_F^A[f(t)] = \begin{cases} \int_{-\infty}^{\infty} f(t) K_A(t, u) dt, & b \neq 0 \\ \sqrt{d} e^{\frac{j}{2c} du^2} f(du), & b = 0 \end{cases} \quad (1)$$

where parameters a, b, c, d are real numbers and satisfying $ad - bc = 1$, and the kernel $K_A(t, u)$ is given by

$$K_A(t, u) = \sqrt{\frac{1}{j2\pi b}} e^{j(\frac{d}{2b}u^2 - \frac{1}{b}ut + \frac{a}{2b}t^2)}$$

When parameter $(a, b, c, d) = (0, 1, -1, 0)$, the LCT coincides with the conventional FM. When parameter $(a, b, c, d) = (\cos \theta, \sin \theta, -\sin \theta, \cos \theta)$, the LCT equals to the fractional FRFT Li et al. (2007). When parameter $(a, b, c, d) = (k, 0, 0, \frac{1}{k})$, the LCT changes into the scaling operation. When parameter $(a, b, c, d) = (1, 0, q, 1)$, the LCT reduces to the chirp multiplication operation. When parameter $(a, b, c, d) = (1, b, 0, 1)$, the LCT reduces to the Fresnel transform (Ozaktas, Kutay, and Zalevsky 2000).

2.2. Wigner distribution and AF

The Wigner distribution of a signal $f(x)$ is defined as (Alieva and Bastiaans 2000)

$$W_f(x, u) = \int_{-\infty}^{\infty} f\left(x + \frac{1}{2}x'\right) f^*\left(x - \frac{1}{2}x'\right) \exp(-j2\pi ux') dx' \quad (2)$$

The AF defined by (Alieva and Bastiaans 2000)

$$A_f(x', u') = \int_{-\infty}^{\infty} f\left(x + \frac{1}{2}x'\right) f^*\left(x - \frac{1}{2}x'\right) \exp(-j2\pi u'x) dx \quad (3)$$

The relationship between the WD and the AF takes the form of a combined forward inverse Fourier transformation (Bastiaans and Alieva 2002)

$$A_f(x', u') = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_f(x, u) \exp(-j2\pi(u'x - ux')) dx du \quad (4)$$

The total energy E of the signal can be derived as following (Bastiaans and Alieva 2002):

$$E = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_f(x, u) dx du = A_f(0, 0) = \int_{-\infty}^{\infty} |f(x)|^2 dx \quad (5)$$

The normalised moment of the WD with $p, q \geq 0$ are defined as follows:

$$\begin{aligned}\mu_{pq}E &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_f(x, u) x^p u^q dx du \\ &= \frac{(-1)^p}{(j2\pi)^{p+q}} \left. \frac{\partial^{p+q} A_f(x, u)}{\partial x^p \partial u^q} \right|_{x=u=0}\end{aligned}\quad (6)$$

3. The moments associate with the LCT

3.1. The relationship between the LCT and wigner distribution

One of the most important properties of the LCT is that it corresponds to a coordinate transformation of the WD and the AF in the phase space (Li and Xu 2012)

$$W_{F_A}(x, u) = W_f(dx - bu, -cx + au)$$

$$A_{F_A}(x', u') = A_f(dx' - bu', -cx' + au')$$

Moreover, the linear canonical power spectra, i.e, the squared modulus of the LCT is related to the WD, and the squared modulus: $|F_A(x)|^2$ of the LCT is directly related to the AF through an inverse Fourier transformation (Bastiaans and Alieva 2002)

$$|F_A(x)|^2 = \int_{-\infty}^{\infty} W_{F_A}(x, u) du$$

$$|F_A(x)|^2 = \int_{-\infty}^{\infty} A_{F_A}(0, u') \exp(j2\pi u' x) du'$$

In the remainder of this article we will concentrate on the WD and its moments.

3.2. Moments in the linear canonical domain

Based on the relationship of the WD and the LCT, the moments associated with the LCT can be expressed by the following theorem.

Theorem 1: Suppose the WD moments of a signal $f(t)$ is $\mu_{p, q}$ in the traditional FM domain, then the moments of this signal in the LCT domain of parameters a, b, c, d can be expressed as the following:

$$v_{pq} = \sum_{k=0}^p \sum_{l=0}^q \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} a^{p-k} b^k c^l d^{q-l} \mu_{p-k+l, q-l+k}$$

Proof: By the definition of the WD moments, the moments associated with the LCT can be written as

$$v_{pq}E = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_{F_A}(x, u) x^p u^q dx du$$

From Equations (4) and (5), we obtain

$$\begin{aligned} v_{pq}E &= \frac{(-1)^p}{(j2\pi)^{p+q}} \frac{\partial^{p+q} A_{(a,b,c,d)}(x,u)}{\partial x^p \partial u^q} \Big|_{x=u=0} \\ &= \frac{(-1)^p}{(j2\pi)^{p+q}} \frac{\partial^{p+q} A_f(dx - bu, -cx + au)}{\partial x^p \partial u^q} \Big|_{x=u=0}. \end{aligned}$$

Let $x_1 = dx - bu$, $u_1 = -cx + au$, then we have

$$\begin{aligned} v_{pq}E &= \frac{(-1)^p}{(j2\pi)^{p+q}} \frac{\partial^{p+q} A_f(x_1, u_1)}{\partial (ax_1 + bu_1)^p \partial (cx_1 + du_1)^q} \Big|_{x_1=u_1=0} \\ &= \frac{(-1)^p}{(j2\pi)^{p+q}} \left(a \frac{\partial}{\partial x_1} + b \frac{\partial}{\partial u_1} \right)^p \left(c \frac{\partial}{\partial x_1} + d \frac{\partial}{\partial u_1} \right)^q A_f(x_1, u_1) \Big|_{x_1=u_1=0} \\ &= \sum_{k=0}^p \sum_{l=0}^q \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} a^{p-k} b^k c^l d^{q-l} \mu_{p-k+l, q-l+k} E \end{aligned}$$

Thus, we have

$$v_{pq} = \sum_{k=0}^p \sum_{l=0}^q \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} a^{p-k} b^k c^l d^{q-l} \mu_{p-k+l, q-l+k}$$

with $p, q \geq 0$.

Corollary 1: Let $q=0$; $p \geq 0$, then we have

$$v_{p0} = \sum_{k=0}^p \begin{bmatrix} p \\ k \end{bmatrix} \mu_{p-k,k} a^{p-k} b^k$$

This situation has been studied in detail in Alieva and Bastiaans (2000).

Corollary 2: Let $A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, then we obtain the same result proposed in Bastiaans and Alieva (2002)

$$v_{pq}A = \sum_{k=0}^p \sum_{l=0}^q \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} (-1)^l \cos \theta^{p+q-k-l} \sin \theta^{k+l} \mu_{p-k+l, q-l+k}$$

with $p, q \geq 0$.

It means that the result derived in this article can be looked as the generalisation of the classical results (Alieva and Bastiaans 2000).

4. The Wigner moments associated with two-dimensional LCT

The two-dimensional LCT with parameters $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ of a signal $f(x, y)$ is defined as follows (James and Agarwal 1996; Ozaktas et al. 2000; Ding and Pei 2011):

$$F_{A,A_1}(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) k_A(x, u) k_{A_1}(y, v) dx dy \quad (7)$$

where the kernel $k_A(x, u)$ $k_{A_1}(y, v)$ is given by

$$k_A(x, u) = \sqrt{\frac{1}{j2\pi b}} e^{j(\frac{d}{2b}u^2 - \frac{1}{b}ux + \frac{a}{2b}x^2)}$$

$$k_{A_1}(y, v) = \sqrt{\frac{1}{j2\pi b_1}} e^{j(\frac{d_1}{2b_1}v^2 - \frac{1}{b_1}vy + \frac{a_1}{2b_1}y^2)}$$

We remark that $F_{(1,0,0,1)(1,0,0,1)}$ represents the signal itself, while $F_{(0,1,-1,0)(0,1,-1,0)}$ corresponds to the normal two-dimensional FT of the signal $f(x, y)$.

The Wigner distribution (WD) of a two-dimensional signal $f(x, y)$ is defined by Bastiaans and Alieva (2002)

$$W_f(x, u; y, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f\left(x + \frac{1}{2}x', y + \frac{1}{2}y'\right) f^*\left(x - \frac{1}{2}x', y - \frac{1}{2}y'\right) \times \exp\{-j2\pi(ux' + vy')\} dx' dy' \quad (8)$$

The AF is defined by (Bastiaans and Alieva 2002)

$$A_f(x', u'; y', v') = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f\left(x + \frac{1}{2}x', y + \frac{1}{2}y'\right) f^*\left(x - \frac{1}{2}x', y - \frac{1}{2}y'\right) \times \exp\{-j2\pi(u'x + v'y)\} dx dy \quad (9)$$

Similarly, the relationship between the two-dimensional WD and the two-dimensional AF is (Bastiaans and Alieva 2002)

$$A_f(x', u'; y', v') = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_f(x, u; y, v) \times \exp\{-j2\pi(u'x - ux' + v'y - vy')\} dx du dy dv \quad (10)$$

The total energy of the signal is

$$E = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_f(x, u; y, v) dx du dy dv$$

$$= A_f(0, 0; 0, 0) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x, y)|^2 dx dy \quad (11)$$

These normalised moments μ_{pqrs} is (Bastiaans and Alieva 2002)

$$\mu_{pqrs} E = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_f(x, u; y, v) x^p u^q y^r v^s dx du dy dv$$

$$= \frac{(-1)^{p+r}}{(j2\pi)^{p+q+r+s}} \frac{\partial^{p+q+r+s} A_f(x, u; y, v)}{\partial x^p \partial u^q \partial y^r \partial v^s} \Big|_{x=u=y=v=0} \quad (12)$$

We can get a promotion in Li and Xu (2012) that the relationship between the classical two-dimensional Wigner distribution (AF) and the two-dimensional linear canonical transform can be rewritten as:

$$W_{F_{AA_1}}(x, u; y, v) = W_f(dx - bu, -cx + au; d_1y - b_1v, -c_1y + a_1v)$$

$$A_{F_{AA_1}}(x', u'; y', v') = W_f(dx' - bu', -cx' + au'; d_1y' - b_1v', -c_1y' + a_1v')$$

Using the above equation, we obtain the moments of the two-dimensional LCT.

Theorem 2: Suppose the WD moments of a signal $f(x, y)$ is μ_{pqrs} associated with the traditional FM domain, then the moments of this signal in the two-dimensional LCT domain of parameters A , A_1 can be expressed as the following:

$$\begin{aligned} v_{pqrs} = & \sum_{k=0}^p \sum_{l=0}^q \sum_{m=0}^r \sum_{n=0}^s \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} \begin{bmatrix} r \\ m \end{bmatrix} \begin{bmatrix} s \\ n \end{bmatrix} \\ & \times a^{p-k} b^k c^l d^{q-l} a_1^{r-m} b_1^m c_1^n d_1^{s-n} \mu_{p-k+l, q-l+k, r-m+n, s-n+m} \end{aligned}$$

Proof: By the definition of the Wigner moments, the moments associated with the two-dimensional LCT can be written as

$$v_{pqrs} E = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} W_{F_{AA_1}}(x, u; y, v) x^p u^q y^r v^s dx du dy dv$$

From Equations (11) and (12), we have

$$\begin{aligned} v_{pqrs} E &= \frac{(-1)^{p+r}}{(j2\pi)^{p+q+r+s}} \frac{\partial^{p+q+r+s} A_{A, A_1}(x, u; y, v)}{\partial x^p \partial u^q \partial y^s \partial v^r} \Big|_{x=u=y=v=0} \\ &= \frac{(-1)^{p+r}}{(j2\pi)^{p+q+r+s}} \frac{\partial^{p+q+r+s} A_f(dx - bu, -cx + au; d_1 y - b_1 v, -c_1 y + a_1 v)}{\partial x^p \partial u^q \partial y^s \partial v^r} \Big|_{x=u=y=v=0} \end{aligned}$$

Let $x_1 = dx - bu$, $u_1 = -cx + au$, $y_1 = d_1 y - b_1 v$, $v_1 = -c_1 y + a_1 v$, then we have

$$\begin{aligned} v_{pqrs} E &= \frac{(-1)^{p+r}}{(j2\pi)^{p+q+r+s}} \frac{\partial^{p+q+r+s} A_f(x_1, u_1, y_1, v_1)}{\partial (ax_1 + bu_1)^p \partial (cx_1 + du_1)^q \partial (a_1 y_1 + b_1 v_1)^p \partial (c_1 y_1 + d_1 v_1)^q} \Big|_{x_1=u_1=y_1=v_1=0} \\ &= \frac{(-1)^{p+r}}{(j2\pi)^{p+q+r+s}} \left(a \frac{\partial}{\partial x_1} + b \frac{\partial}{\partial u_1} \right)^p \left(c \frac{\partial}{\partial x_1} + d \frac{\partial}{\partial u_1} \right)^q \\ &\quad \times \left(a_1 \frac{\partial}{\partial y_1} + b_1 \frac{\partial}{\partial v_1} \right)^r \left(c_1 \frac{\partial}{\partial y_1} + d_1 \frac{\partial}{\partial v_1} \right)^s A_f(x_1, u_1; y_1, v_1) \Big|_{x_1=u_1=y_1=v_1=0} \\ &= \sum_{k=0}^p \sum_{l=0}^q \sum_{m=0}^r \sum_{n=0}^s \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} \begin{bmatrix} r \\ m \end{bmatrix} \begin{bmatrix} s \\ n \end{bmatrix} a^{p-k} b^k c^l d^{q-l} a_1^{r-m} b_1^m c_1^n d_1^{s-n} \\ &\quad \times \mu_{p-k+l, q-l+k, r-m+n, s-n+m} E \end{aligned}$$

Thus, we have

$$v_{pqrs} = \sum_{k=0}^p \sum_{l=0}^q \sum_{m=0}^r \sum_{n=0}^s \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} \begin{bmatrix} r \\ m \end{bmatrix} \begin{bmatrix} s \\ n \end{bmatrix} a^{p-k} b^k c^l d^{q-l} a_1^{r-m} b_1^m c_1^n d_1^{s-n} \mu_{p-k+l, q-l+k, r-m+n, s-n+m}$$

with $p, q, r, s \geq 0$

Corollary 3: In particular, for $q = s = 0$; $p, r \geq 0$ we have

$$v_{p0r0} = \sum_{k=0}^p \sum_{m=0}^r \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} r \\ m \end{bmatrix} a^{p-k} b^k a_1^{r-m} b_1^m \mu_{p-k, q-k, r-m, m} \quad (13)$$

This situation has been studied in Bastiaans and Alieva (2002) in detail.

Corollary 4: Considering $(a, b, c, d) = (\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha)$, $(a_1, b_1, c_1, d_1) = (\cos \beta, \sin \beta, -\sin \beta, \cos \beta)$, we have (Bastiaans and Alieva 2002)

$$\begin{aligned} v_{pqrs} = & \sum_{k=0}^p \sum_{l=0}^q \sum_{m=0}^r \sum_{n=0}^s \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} \begin{bmatrix} r \\ m \end{bmatrix} \begin{bmatrix} s \\ n \end{bmatrix} (-1)^{l+n} \cos \alpha^{p-k+q-l} \sin \alpha^{k+l} \\ & \times \cos \beta^{r-m+s-n} \sin \beta^{m+n} \mu_{p-k+l, q-l+k, r-m+n, s-n+m} \end{aligned}$$

5. Simulation experiments

In order to support the conclusions of this article, the simulations experiments for the one-dimensional cases are discussed in this section. Let $q=0$ of the one-dimensional form

$$v_{pq} = \sum_{k=0}^p \sum_{l=0}^q \begin{bmatrix} p \\ k \end{bmatrix} \begin{bmatrix} q \\ l \end{bmatrix} a^{p-k} b^k c^l d^{q-l} \mu_{p-k+l, q-l+k}$$

Then, we obtain

$$v_{p0} = \sum_{k=0}^p \begin{bmatrix} p \\ k \end{bmatrix} a^{p-k} b^k \mu_{p-k, k}.$$

We choose the signal to be $f(t) = 2e^{-j(t+2t^2)}$ and let the parameter of the LCT to be $a=2$, $b=1$, $c=0$, $d=0.5$, we use v'_{p0} to express $\sum_{k=0}^p \begin{bmatrix} p \\ k \end{bmatrix} a^{p-k} b^k \mu_{p-k, k}$ and use ε to express the difference between v_{p0} and v'_{p0} . The following results can be obtained for different values of p , it is shown in these results that the derived theories of this article is correct. The two-dimensional cases can also be verified by the similar methods in simulation.

p	v_{p0}	v'_{p0}	ε
0	0.0314	-0.0056	0.0370
1	0.1272	-0.0405	0.1677
2	-0.0004	0.0052	-0.0054
3	0.0048	0.0096	-0.0048
4	-0.0001	0.0084	-0.0085
5	0.0002	0.0430	-0.0432
6	-0.0000	0.0166	-0.0166
7	0.0000	0.0953	-0.0953
8	-0.0000	0.0360	-0.0360

6. Conclusions

Focussing on the Wigner distribution moments associated with the LCT, this article investigates the moments problems based on the LCT. First, we use the moments of the Wigner distribution and the relationship between the Wigner distribution and the LCT to deduce the moments associated with the LCT. Then, we extend this theory into the two-dimensional cases. The simulation experiment are also proposed to verify the derived results. In future works, we will discuss the properties of the moments associated with the LCT in detail and its applications in signal-processing fields.

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References

- Abu-Mostafa, Y.S., and Psaltis, D. (1984), 'Recognitive Aspects of Moment Invariants', *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 6, 698–706.
- Alieva, T., and Bastiaans, M.J. (2000), 'On Fractional Fourier Transform Moments', *IEEE Signal Processing Letters*, 7, 320–323.
- Bai, R.-F., Li, B.-Z., and Cheng, Q.-Y. (2012), 'Wigner–Ville Distribution Associated with the Linear Canonical Transform', *Journal of Applied Mathematics*, 2012, 1–11.
- Bastiaans, M.J., and Alieva, T. (2002), 'Wigner Distribution Moments in Fractional Fourier Transform Systems', *Journal of the Optical Society of America*, 19, 1763–1773.
- Belkasim, S.O., Shridhar, M., and Ahmadi, M. (1991), 'Pattern Recognition with Moment Invariants: A Comparative Study and New Results', *Pattern Recognition*, 24, 1117–1138.
- Che, T.-W., Li, B.-Z., and Xu, T.-Z. (2012), 'The Ambiguity Function Associated with the Linear Canonical Transform', *EURASIP Journal on Advances in Signal Processing*, 138, 1–14.
- Ding, J.-J., and Pei, S.-C. (2011), 'Eigenfunctions and Self-imaging Phenomena of the Two-dimensional Nonseparable Linear Canonical Transform', *Journal of the Optical Society of America*, 28, 82–95.
- Flusser, J., and Suk, T. (1993), 'Pattern Recognition by Affine Moment Invariants', *Pattern Recognition*, 26, 167–174.
- Hu, M.K. (1962), 'Visual Pattern Recognition by Moment Invariants', *IRE Transactions on Information Theory*, 8, 179–187.
- James, D.F.V., and Agarwal, G.S. (1996), 'The Generalised Fresnel Transform and its Applications to Optics', *Optics Communications*, 126, 207–212.
- Li, B.-Z., Tao, R., and Wang, Y. (2007), 'New Sampling Formulae Related to Linear Canonical Transform', *Signal Processing*, 87, 983–990.
- Li, B.-Z., and Xu, T.-Z. (2012), 'Spectral Analysis of Sampled Signals in the Linear Canonical Transform Domain', *Mathematical Problems in Engineering*, 2012, 19.
- Liu, S.T., and Tsai, W.H. (1990), 'Moment-preserving Corner Detection', *Pattern Recognition*, 23, 441–446.
- Lo, C.H., and Don, H.S. (1989), '3-D Moment Forms: Their Construction and Application to Object Identification and Positioning', *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11, 1053–1064.
- Luo, L.M., Hamitouche, C., Dillenseger, J.L., and Coatrieux, J.L. (1993), 'A Moment-based Three-dimensional Edge Operator', *IEEE Transactions on Biomedical Engineering*, 40, 693–703.
- Moshinsky, M., and Quesne, C. (1971), 'Linear Canonical Transformations and their Unitary Representations', *Journal of Mathematical Physics*, 12, 1772–1783.
- Ozaktas, H.M., Kutay, M.A., and Zalevsky, Z. (2000), *The Fractional Fourier Transform with Applications in Optics and Signal Processing*, New York: Wiley.
- Tao, R., Li, B.-Z., and Wang, Y. (2008), 'On Sampling of Bandlimited Signals Associated with the Linear Canonical Transform', *IEEE Transactions on Signal Processing*, 56, 5454–5464.
- Teh, C.H., and Chin, R.T. (1988), 'On Image Analysis by the Methods of Moments', *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 10, 496–513.
- Teague, M.R. (1980), 'Image Analysis via the General Theory of Moments', *Journal of the Optical Society of America*, 70, 920–930.
- Tuceryan, M. (1994), 'Moment-based Texture Segmentation', *Pattern Recognition Letters*, 15, 659–668.

- Wei, D., Ran, Q., and Li, Y. (2011), 'Reconstruction of Band-limited Signals from Multichannel and Periodic Non-Uniform Samples in the Linear Canonical Transform Domain', *Optics Communications*, 284, 4307–4315.
- Yokoya, N., and Levine, M.D. (1989), 'Range Image Segmentation Based on Differential Geometry: A Hybrid Approach', *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11, 643–649.